

Improving Access to Healthcare by Removing Non-Medical Barriers
Team 2 - Machine Learning Learning Machines
Christine Le, Jason Leff, Tiffany Le, Brendan Phoebe, and Chloe Florence

Caring about the patient's journey is about caring about all patients' journeys. Healthcare access is essential for general wellness and treating chronic conditions in non-urban areas. We sought to improve access to healthcare by addressing non-medical barriers such as patient residential distance from care clinics and increasing the agency patients have over their medical journeys. The data was structured as a relational database so SQL was utilized to partition data into subsets to fulfill analytics needs.

Stormont Vail Hospital (SVH) is the busiest hospital in Kansas and the only ACS Level II Trauma center in the region. We analysed the effect of census tract – or place of residency – on how often patients encountered the Stormont Vail medical system. Mean annual patient visits to SVH are heavily-right skewed, exhibiting distinctly high values in the Northeast and Flint Hills regions of the state and low values in the Southwest. Segmented Poisson Regression reveals a piecewise log-linear relationship between the distance of patient residencies to Stormont Vail Hospital and mean annual patient visits. The decrease in annual visits per mile is estimated to change when patient residencies exceed 21.09 miles (with a standard error 0.495) from SVH. Visits from patients who live 21.09 miles or closer from SVH decrease by 2.16% ($p < 2e-16$) per additional mile from SVH. However, visits are approximately constant beyond this threshold. We hypothesize that patients who live close to SVH may visit the facility for routine care, whereas patients that live far from SVH visit at a constant rate in proportion to emergencies and routinely visit other care centers.

MyChart is an online platform that allows patients to track medical documents, schedule appointments, and connect to medical professionals. We explored the relationship between MyChart users and the number of encounters patients have with Stormont Vail. This information is critical for understanding and improving services for chronic conditions for which multiple visits to Stormont Vail facilities are expected. We found that, on average, there is a statistically significant difference in appointments between patients who created and activated a MyChart account and those who did not. We are 95% confident that the patients with MyChart had a true increase of 11.15 to 13.36 appointments during the four-year span of the study compared to non-MyChart users. Increasing online scheduling accessibility is especially important for patients with chronic issues who need numerous appointments to manage their healthcare.

Combining the previous two findings, we determined an interaction effect exists between distance from SVH and the status of patient MyChart accounts. We applied a log-transformation to mean annual patient visits to dampen the effect of the heavy skew. We fit a linear model and found statistical significance in interaction and predictor effects ($p < 2e-16$) for these two predictors while passing other basic diagnostic checks in constant variance and outliers. Understanding the relationship between proximity to healthcare facilities and online access is essential to creating the best possible journey for every Stormont Vail patient.